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New Paleomagnetic Data From the Antarctic Peninsula and Their Tectonic Implications

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New paleomagnetic data presented here further constrain the relative motion of the Antarctic Peninsula relative to East Antarctica during the Mesozoic development of the southern ocean basins. The Antarctic Peninsula (AP) is one of four crustal blocks that define West Antarctica, the others being the Ellsworth-Whitmore Mountains (EWM), Thurston Island-Eights Coast (TI), and Marie Byrd Land (MBL). A Jurassic pole (~155Ma) (124°E, 64°S, $A95$=7.1°, N=10 VGPs) was obtained from the AP block which suggests that the AP block rotated clockwise between ~175 and ~155 Ma due to significant early opening in the Weddell Sea basin. A new Early Cretaceous paleomagnetic pole (182°E, 74S, $A95$=5.9°, N=6 site mean VGPs) indicates that the AP block was in or near to its present-day position with respect to East Antarctica by ~130 Ma. Between ~155 and 130 Ma, counterclockwise rotation of the AP-TI blocks, together with the southward motion of East Gondwanaland, probably resulted in subduction of Weddell Sea ocean floor beneath the southern AP block and initiated the Palmer Land deformational event. A ~130 Ma pole from the TI block requires clockwise rotation of the TI and possibly the EWM blocks between 130 and ~110 Ma producing sinsistral strike-slip motion between the EWM block and East Antarctica and dextral transpressional motion between the TI-EWM blocks and the AP block. New AP block ~110 Ma and ~85 Ma poles from this study (199°E, 74°S, $A95$=6.9°, N=13 VGPs; 152°E, 86°S, $A95$= 7.5°, N=6 VGPs, respectively) are similar to equivalent age poles from East Antarctica and suggest little or no relative motion between the Antarctic Peninsula and East Antarctica. Northern and southern Antarctic Peninsula mid-Cretaceous poles are very much alike suggesting that the "S" shape of the Antarctic Peninsula is not due to oroclinal bending since ~110 Ma.

INTRODUCTION

This paper presents new paleomagnetic data from the Antarctic Peninsula that when combined with existing geologic and geophysical data help to constrain the motions of the Antarctic Peninsula relative to East Antarctica and to other crustal blocks in West Antarctica. The location of West Antarctica with respect to cratonic East Antarctica prior to and after the Middle Jurassic breakup of the Gondwanaland supercontinent is important for understanding the opening history of the Weddell Sea. West Antarctica is composed of four major crustal blocks: the Antarctic Peninsula (AP), the Ellsworth-Whitmore Mountains (EWM), the Thurston Island-Eights Coast (TI), and Marie Byrd Land (MBL) [*Jankowski and Drewry*, 1981; *Dalziel and Elliot*, 1982] (Figure 1a). The relationship of these blocks to each other and to East Antarctica is not well constrained prior to the Cenozoic. The new paleomagnetic data provide some constraint on the pre-Cenozoic relative positions of these crustal blocks as well as leading to a better understanding of the opening history of the Weddell Sea.

GEOLOGIC OVERVIEW OF THE ANTARCTIC PENINSULA

The AP block consists of the area between ~280° E and ~310° E and from 60°S to ~ 76°S (Figure 1a). The majority of rocks from the S-shaped AP block are igneous (both plutonic and volcanic rocks) with lesser occurrences of sedimentary and metamorphic rocks. Apart from subduction complex rocks [*Grunow et al.*, 1992] there are no Mesozoic or Cenozoic rocks considered to be parts of exotic terranes accreted to the Antarctic Peninsula. Indeed, there are no obviously exotic terranes in the West Antarctic blocks from at least the Mesozoic onward.

Most of the rocks along the northern Antarctic Peninsula (Graham Land) are mid-Mesozoic to Cenozoic arc rocks associated with subduction of the Phoenix (Aluk) plate beneath the Antarctic Peninsula [*Saunders et al.*, 1982; *Pankhurst*, 1990]. Opening of Drake Passage caused separation of the northern Antarctic Peninsula from southern South America in the Miocene [*Barker and Burrell*, 1977]. The volcanically active marginal basin of Bransfield Strait has been opening for the past 1.5 to 2 m.y. [*Barker*, 1982] separating the South Shetland Islands from the mainland of the Antarctic Peninsula (Figure 1a).

The physical connection of the AP block to the other West Antarctic blocks is hidden by ice. *Ford* [1972] suggests that the boundary between the AP and TI blocks may have been a Late Cretaceous strike-slip fault. The boundary between the AP and the EWM block is hidden by the grabenlike structure of the Evans Ice Stream (Figure 1a) [*Garrett et al.*, 1987].

PALEOMAGNETIC RESULTS

New paleomagnetic data are presented from igneous and sedimentary rocks collected from the northern Antarctic Peninsula and are described by region, starting with results from the youngest rocks (see Figure 1b for locations). Most of the rocks exposed in the AP block are intrusions and an attempt was made to collect samples from as many widely separated sites and different structural/cooling units as possible within any given intrusion in order to average out secular variation and evaluate the possibility of tectonic tilting. Generally six to seven cores were sampled at each site (four to five samples in dikes of less than 1 m width) with a portable

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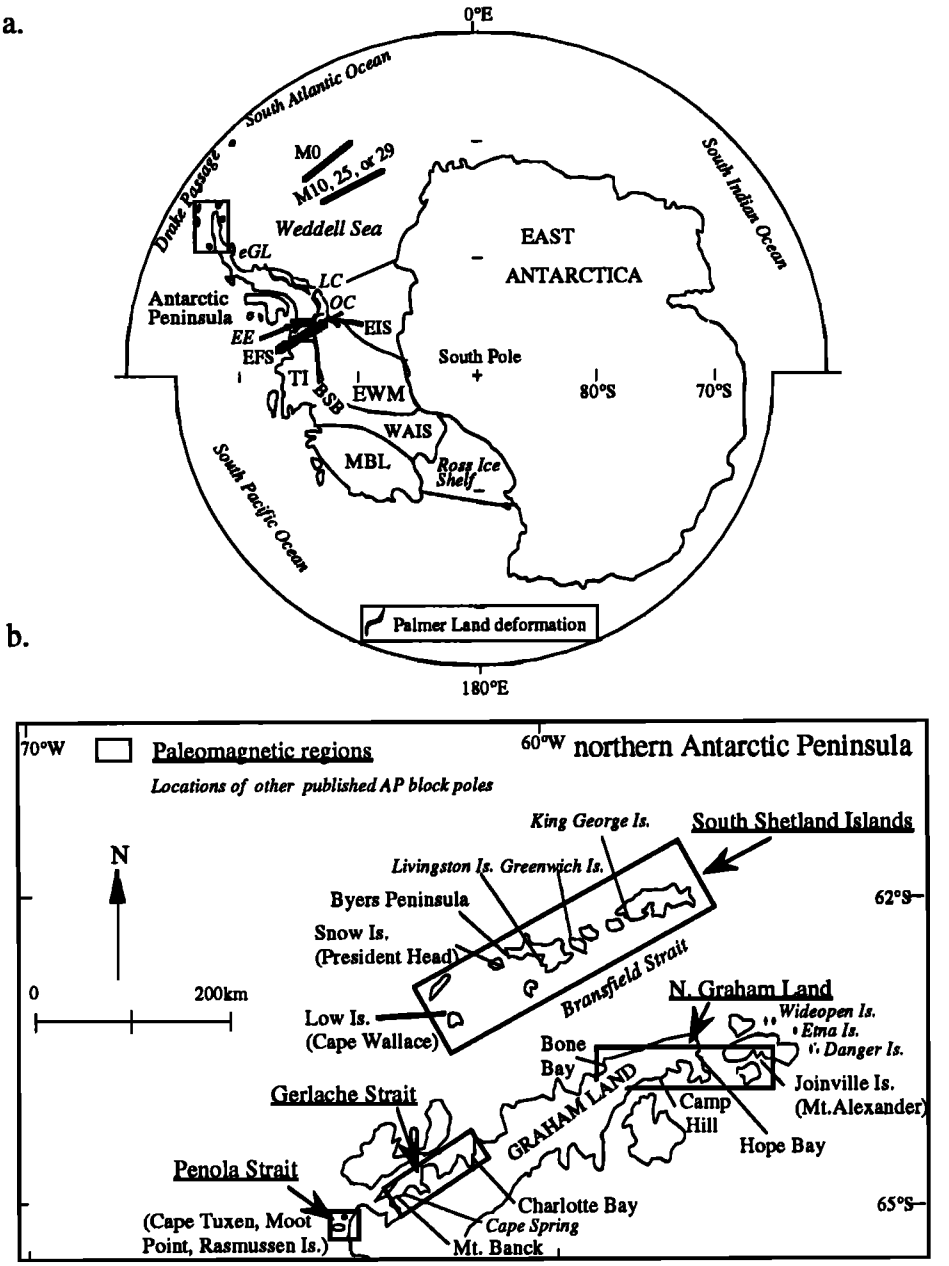


Fig. 1. (a) The West Antarctic crustal blocks in their present-day positions with respect to East Antarctica. BSB, Byrd Subglacial Basin; EE, Eastern Ellsworth land; EFS, Ellsworth Fault system; EIS, Evans Ice Stream; EWM, Ellsworth-Whitmore Mountains block; eGL, east Graham Land; LC, Lassiter Coast; MBL, Marie Byrd Land; OC, Orville Coast; TI, Thurston Island block; WAIS, West Antarctic ice sheet. (b) Sample locations from the northern Antarctic Peninsula.

Pomeroy rock coring drill and oriented using a Brunton compass. In some locations, time constraints and poor weather conditions prevented us from collecting at more than two or three sites. Sun compass measurements (or sightings) were made at each locality to allow correction for local magnetic variation and were within 3° of the predicted magnetic variation.

The samples were analyzed at the the University of Oxford and the Lamont-Doherty Earth Observatory of Columbia University paleomagnetism laboratories. Measurements of the natural remanent magnetization (NRM) were made using a cryogenic magnetometer or a computerized fluxgate Molspin magnetometer on one or more specimens cut from each core. Pilot samples were progressively demagnetized using step-wise alternating field (AF) or thermal (TH) techniques to determine the best demagnetization procedure for each site. Steps of generally 5 to 10 mT up to a peak of 95 mT were used during AF demagnetization, while steps of generally 50°C up to 500°C and then smaller steps up to a maximum of 660°C were used in thermal demagnetization. Susceptibility measurements were made after each thermal demagnetization step to evaluate mineralogic changes. Isothermal remanent magnetization (IRM) experiments were carried out on representative speciments from the main sample locations. Magnetic saturation was obtained in fields of less than 300 mT for the igneous rocks and when combined with the observed unblocking temperatures suggests that the remanence is carried primarily by titanomagnetite.

The demagnetization data were plotted on vector endpoint diagrams, and characteristic magnetization directions were

calculated for linear segments converging toward the origin using principal component analysis [*Kirschvink*, 1980]. Directions were considered meaningful if defined by three or more colinear demagnetization steps with maximum angular deviations of less than 10° (they were generally less than 5°). Mean directions for each locality were obtained by combining sample means to site means and then site means to a unit mean using standard *Fisher* [1953] statistics. Only site mean directions with $\alpha 95s \leq 15°$ were used for calculating site mean virtual geomagnetic poles (VGPs). Location mean poles were calculated by using site-mean VGPs unless otherwise stated.

Radiometric age control on the igneous rocks is based on previously published data and on new U-Pb zircon results collected from the paleomagnetic sampling sites by S.B. Mukasa and J. Tangeman. Baked contacts of dikes cutting plutonic rocks always showed similar directions to the plutonic rocks which constitutes a positive baked-contact test, but it is likely that in many of these cases the dikes are comagmatic with the plutons they intrude.

Penola Strait

Cape Tuxen. A coarse-grained, hornblende ± biotite bearing banded gabbro crops out on a prominent point at Cape Tuxen [*Curtis*, 1966]. U-Pb dating of zircons has yielded an age of 85 ± 1 Ma for the gabbro(J. Tangeman and S.Mukasa,personal communication, 1992). Three sites were drilled in the gabbro. All of the sites occur more than 50 m west of a northeast striking, vertical fault with subhorizontal slickensides [*Curtis*, 1966]. A single component of magnetization pointing steeply upward and to the north was found in the gabbro sites (Table 1 and Figure 2a). No correction has been made for the nearly vertical magmatic banding since it produces unrealistically low (-5°) inclinations.

One kilometer to the southeast of Cape Tuxen, there is a granodiorite body which has been dated at 84.5 ± 1 Ma using the U-Pb method on zircons (J.Tangeman and S.Mukasa,personal communication, 1992). Dikes of the granodiorite intruding the gabbro were observed on cliff faces. The paleomagnetic locations in granodiorite occur on the opposite side of the above mentioned strike-slip fault in the gabbro. Three sites were drilled in this rock and yielded directions that were steeply upward to the north. The unit mean direction for the six sites from the gabbro and granodiorite is D= 4.7°; I= -75.2° (Table 2 and Figure 2a) similar to the NRM directions (D= 9°; I= -77°) observed by *Blundell* [1962] from Cape Tuxen and neighboring islands.

Moot Point. Moot Point forms a series of low islands and a promontory along Penola Strait and consists of medium grained, biotite ± hornblende bearing diorite cut by nearly vertical north-south trending mafic dikes of which two were sampled for baked contact with the diorite. Four sites were drilled in the Moot Point diorite (U-Pb zircon age of 106 ± 1 Ma,(J.Tangeman and S.Mukasa,personal communication, 1992)). Although no age determinations have been made on the mafic dikes they yield directions indistinguishable from the diorite (Table 1 and Figure 2b) and were not observed cutting the Cape Tuxen rocks which suggests that the dikes are older than ~85 Ma and are possibly comagmatic with the diorite.

Two distinct components (A and B) of magnetization were determined from the diorite but not from the mafic dikes. The A component site mean directions from the diorite and a mafic dike (one dike was magnetically unstable) are indistinguishable (Table 1 and Figure 2b) and point upward and to the north-northeast; the mean direction for all five sites is D= 23.8°, I= -71.2° (Table 2and Figure 2b). In some diorite samples, a second, B component of magnetization (Table 2; D= 297.2°, I= -76.3°, n=13 samples) was found pointing northwest and upward (with one B component sample pointing southeast and downward) (Figure 2b).

Rasmussen Island. The outcrops at Rasmussen Island (an islet and a headland) consist of fractured, coarse-grained, pink, biotite-bearing granite cut by mafic dikes. The granite has an age of 117 Ma ± 1 (J. tangeman and S. Mukasa, personal communication, 1992). Four sites were drilled in the granite and one in a mafic dike.

A single component of magnetization, that was up and to the northeast (Table 1; Figure 2c) was observed in both the granite and the mafic dike after thermal demagnetization. The unit mean direction for the five sites is D= 14.4°, I= -72.1° (Table 2 and Figure 2c).

Gerlache Strait

Charlotte Bay. The southeastern end of Charlotte Bay consists of a gabbro cut by a porphyritic coarse-grained, quartz monzonite (which has yielded a U-Pb zircon age of 115 ± 8 Ma (J. Tangeman and S. Mukasa, personal communication, 1992) which in turn is cut by mafic dikes. Part of the gabbro occurs as large blocky xenoliths within a finer grained granodioritic host. The contact of the gabbro with the quartz monzonite was not observed. Five sites were drilled at Charlotte Bay: one in the gabbroic xenoliths, one in the granodioritic host rock containing the gabbro xenoliths, one in the porphyritic quartz monzonite, one in the gabbro and one in a north-south trending mafic dike. Baked contact tests were performed on the different units but no significant difference was found in the directions between the different lithologies (Table 1). A single component of magnetization, defined by an upward pointing, northwest directed vector was found in all three rock types (Table 1 and Figure 2d). A second component of magnetization was also observed but was not consistent between samples from the same sites. The unit mean direction for the five sites is D= 349.3°, I= -77.1° (Table 2 and Figure 2d).

Mount Banck. This locality consists of a pink, coarse-grained, biotite-bearing granite (131 ± 4 Ma, Rb-Sr whole rock isochron age [*Pankhurst*, 1982]) intruded at high level and cut by mafic dikes. Three sites were obtained from this largely inaccessible granite.

Two components of magnetization (A and B) were observed in the granite samples after thermal demagnetization. Component A, with generally lower unblocking temperature (Tub <450°C), was directed up and to the northeast (D= 21.6°, I = -70.4°, N=3 sites), while the component B of magnetization (Tub< 570°C) was directed up and to the northwest (D= 327.1°, I= -77.6°, N=10 samples) (Tables 1 and 2 and Figure 2e).

Northern Graham Land

Hope Bay. The Mount Flora Formation at Hope Bay consists of slightly tilted (but without apparent folding) black siltstones, shales, and tan/brown sandstones and conglomerates [*Elliot and Gracanin*, 1983]. The baked rocks contain abundant pyritized fossils. The flora and fauna constrain the age of the sedimentary rocks to Lower Jurassic [*Rees*, 1993]. The Mount Flora Formation is overlain by the

TABLE 1. Antarctic Peninsula Site Mean Directions and Virtual Geomagnetic Poles Determined From This Study

Site	Component	Lithology	Bedding/ Dike Trend	NRM x 10^{-2} A/m	Samples n/N	*D*, deg	*I*, deg	$\alpha 95$	*k*	VGP Longitude °E	VGP Latitude °S
			Penola Strait: Cape Tuxen/Deliverance Point (295.987°E, 65.27°S)								
LM6A		gabbro		1111.40	7\7	5.8	-72.7	10.0	38	139.1	82.3
LM6B		gabbro		75.52	7\7	334.8	-80.5	4.6	171	337.9	80.0
LM6C		gabbro		228.40	4\5	1.0	-72.1	2.4	1427	119.5	81.8
LM7A		granodiorite		245.90	7\7	2.0	-73.0	5.7	112	124.9	83.2
LM7B		granodiorite		15.43	6\6	2.4	-74.4	10.1	45	130.5	85.4
LM7C		granodiorite		54.50	5\6	32.1	-75.9	7.5	105	212.9	76.1
			Penola Strait: Moot Point (295.9°E, 65.2°S)								
LM4A	(Com A)	granodiorite		52.73	6\6	4.0	-73.8	11.5	35	136.9	84.4
	(Com B)				3\6	286.3	-71.9	14.1	78	3.9	55.5
LM4B	(Com A)	granodiorite		45.73	7\9	31.9	-70.3	11.1	30	186.6	71.0
	(Com B)				3\6	332.9	-79.0	19.8	40	355.3	78.9
LM4C		mafic dike	197/90	50.61	5\6	33.6	-70.7	7.7	99	199.9	70.7
LM4D	(Com A)	granodiorite		80.95	7\7	27.6	-70.5	3.4	322	181.4	72.9
	(Com B)				4\7	301.1	-73.7	16.0	34	9.6	63.3
LM4E	(Com A)	granodiorite		58.61	7\7	18.8	-69.1	5.2	137	161.9	74.2
	(Com B)				3\7	275.3	-78.2	10.1	151	343.2	58.5
			Penola Strait: Rasmussen Island (295.92°E, 65.25°S)								
LM5A		granite		20.58	7\7	42.0	-80.0	8.8	48	243.3	73.7
LM5B		granite		23.55	6\6	1.2	-74.3	6.8	98	123.2	85.4
LM5C		granite		14.93	6\6	28.7	-69.0	6.1	122	178.3	70.7
LM5C		mafic dike		280.40	4\4	355.5	-64.5	6.3	215	106.4	70.9
LM5D		granite		34.06	4\4	19.5	-69.2	14.1	52	138.1	74.5
			Gerlache Strait: Charlotte Bay (298.53°E, 64.62°S)								
CB4A		monzonite		46.00	3\3	340.1	-77.0	5.7	469	15.9	81.6
CB4A		gabbroic xenoliths		98.32	5\5	351.2	-74.1	3.0	632	71.3	84.1
CB4A		gabbro		213.04	6\6	334.5	-77.6	4.5	220	8.4	79.4
CB4B		monzonite		28.90	8\8	349.6	-78.6	4.6	145	344.1	84.6
CB4C		mafic dike	203/67W	9.17	4\4	9.9	-76.7	4.9	357	213.4	85.7
			Gerlache Strait: Mt. Banck (296.95°E, 64.9°S)								
BK1A	(Com A)	granite		3.85	6\6	24.8	-71.5	8.2	68	183.0	75.2
	(Com B)				3\6	338.5	-78.4	6.4	369	0.0	81.0
BK1B	(Com A)	granite		2.39	6\6	29.6	-72.1	7.5	81	192.0	73.9
	(Com B)				2\6	354.7	-72.5	--	687	--	--
BK2A	(Com A)	granite		4.68	5\5	12.6	-67.1	14.7	28	146.7	73.6
	(Com B)				5\5	329.4	-75.6	6.8	129	21.5	76.5
			Northern Graham Land: Hope Bay (303°E, 63.38°S)								
HP1A		siltstone	125/32W	0.43	9\9	354.0	-81.1	7.1	53	313.8	80.4
						[26.0	-51.0	7.1	54	162.8	54.3]
HP1B		siltstone	112/15W	0.54	5\5	20.0	-78.3	8.5	82	248.1	80.8
						[21.2	-63.3	8.5	82	166.1	67.9]
HP1C		siltstone	109/20W	5.44	3\5	30.1	-81.3	15.5	64	266.8	75.6
						[22.7	-61.4	15.6	64	165.9	65.3]
HP1D		siltstone	055/13E	0.66	6\6	68.6	-85.2	4.5	224	281.5	65.3
						[346.7	-77.2	4.5	222	5.7	83.9]
HP1E		siltstone	116/39W	0.85	4\5	144.9	-81.9	28.3	12	288.9	49.3
						[34.5	-52.5	21.5	19	174.9	52.9]
HP2A		siltstone	201/30W	8.46	5\5	6.9	-70.7	12.7	37	148.8	80.9
						[72.6	-59.1	12.7	37	219.4	42.6]
HP2B		volcaniclastic rock	168/19W	20.80	6\6	10.8	-76.5	9.6	50	229.9	85.2
						[50.0	-62.9	9.6	50	203.0	56.2]
			Northern Graham Land: Joinville Island (304.2°E, 63.3°S)								
JV1B		dike		180.00	3\3	6.7	-72.4	23.7	28	157.4	83.4
JV2A		gabbro		30.00	6\7	0.6	-78.7	5.9	131	301.4	85.2
JV2B		gabbro (near contact)		36.96	3\7	26.3	-77.4	12.1	105	239.1	78.5
			Northern Graham Land: Bone Bay (300.93°E, 63.63°S)								
BB1A		granite		0.20	6\6	1.9	-56.8	10.4	42	124.3	63.7
BB1B		granite		0.07	7\7	15.0	-54.2	10.8	27	145.9	59.7
BB1B		mafic dike	043/80E	51.30	5\5	351.9	-64.1	4.8	259	102.7	71.7
BB1C		granite		0.09	6\6	26.1	-59.6	13.4	26	166.8	62.2
BB1D		granite		0.04	5\6	359.3	-50.0	9.7	63	119.8	57.2
BB1E		granite		0.16	6\6	10.3	-54.2	9.4	51	138.2	60.4
BB1F		granite		0.05	4\7	40.7	-67.1	18.2	26	197.7	64.4
BB1L		granite		8.78	4\4	344.3	-58.6	9.8	89	92.4	63.9
BB1L		mafic dike		77.04	8\8	350.0	-55.7	5.7	96	103.6	62.0
			South Shetland Islands: Snow Island, President Head (298.8°E, 62.73°S)								
SN1A		volcanic	340/22E	72.76	7\7	316.7	-76.0	1.9	982	7.0	10.8
						[275.9	-59.8	1.9	997	11.9	38.0]
SN1B		volcanic (hematitic)	264/25W	187.46	6\6	349.2	-78.6	4.4	230	334.3	83.1
						[177.5	-76.3	4.4	231	297.4	36.8]
SN1C		siltstone	261/17W	15.53	8\8	359.6	-76.0	4.9	131	312.0	89.2
						[136.3	-86.2	4.9	132	289.2	56.8]

TABLE 1. (continued)

Site	Component	Lithology	Bedding/ Dike Trend	NRM x 10^{-2} A/m	Samples n/N	*D*, deg	*I*, deg	$\alpha 95$	*k*	VGP Longitude °E	VGP Latitude °S
			South Shetland Islands: Low Island, Cape Wallace (297.75°E, 63.22°S)								
LW1A		sandstone	(n: 1-4) 054/23E	0.25	10\10	161.8	69.9	7.5	42	64.3	76.8
			(n: 5-10) 116/20W			[169.8	51.1	11.7	18	101.3	57.9]
LW1B		sandstone/	(n: 1-8) 332/38E	0.25	15\15	220.3	32.8	4.5	72	168.2	36.9
		siltstone	(n: 9-15) 345/34E	0.24		[194.6	61.4	5.1	58	146.9	67.6]
LW1C		volcanic		4.77	7\7	169.5	37.1	3.6	278	103.3	46.9
						[165.9	15.3	3.6	277	100.9	33.6]
			South Shetland Islands: Byers Peninsula, Point Smellie (298.85°E, 62.65°S)								
B1A		lava flow	021/24E	95.53	6\6	358.9	-77.5	2.0	1078	306.2	86.5
						[313.7	-59.3	2.1	1060	46.3	54.6]
B1B		lava flow	024/14E	3.05	5\5	328.4	-77.3	3.5	470	2.0	76.1
						[310.6	-64.5	3.5	467	36.0	58.1]
B1C		lava flow	150/14W	14.08	3\5	322.7	-40.8	20.9	36	68.8	43.4
						[334.9	-41.0	20.9	36	83.8	47.4]
B1C		siltstone	150/14W	0.46	2\2	0.6	-70.9	--	337	121.5	82.7
						[24.0	-61.2		338	164.9	65.3]
B1D		lava flow	264/16W	8.04	3\3	331.3	-47.5	14.6	72	76.6	51.2
						[320.8	-61.8	14.6	72	51.3	60.0]
B1D		siltstone	264/16W	0.23	3\3	13.4	-51.8	19.1	43	140.8	58.6
						[25.0	-66.3	19.1	43	175.0	70.4]
B1E		siltstone	120/13E	0.76	6\6	8.8	-86.0	4.6	210	295.3	70.4
						[24.7	-73.3	4.6	211	202.9	77.5]
B1F		lava flow	091/25E	77.06	7\7	94.6	78.2	6.6	84	162,9	56.5
						[155.1	61.8	6.7	83	70.7	65.5]
			South Shetland Islands: Byers Peninsula, Chester Cone (298.92°E, 62.63°S)								
B2A		diabase plug (Usnea Plug)		12.08	6\7	353.8	-76.5	13.0	28	353.7	86.7
B2B		andesite plug		105.28	5\6	6.5	-65.5	7.1	117	135.6	74.5
B2C		andesite plug		39.46	5\5	8.8	-64.5	11.6	44	140.1	73.0
			South Shetland Islands: Byers Peninsula, Victor Rock Area (298.9°E, 62.68°S)								
B3A		lava flow	330/20E	346.62	4\5	30.0	-72.7	5.2	309	205.2	74.6
						[298.1	-80.2	5.2	308	306.3	55.7]
B3B		lava flow	330/20E	79.68	6\6	26.4	-65.5	5.9	130	175.6	69.0
						[331.1	-76.9	5.9	129	5.7	78.0]
B3C		lava flow	330/24E	334.96	6\6	23.1	-80.8	8.3	66	265.9	77.2
						[263.2	-76.3	8.3	66	342.4	50.8]
			South Shetland Islands: Byers Peninsula, Laager Point (298.85°E, 62.63°S)								
B4A		siltstone	268/30W	0.04	6\6	356.1	-67.4	11.5	35	107.2	77.4
						168.5	-85.0	10.7	40	295.6	52.9]
B4B		siltstone	289/26E	0.78	6\6	11.0	-65.8	4.3	241	146.7	74.2
						[253.8	-82.7	4.9	186	324.0	56.0]
B4C		siltstone	294/33E	0.36	4\4	21.8	-68.6	12.6	54	176.3	74.3
						[204.1	-78.7	12.4	56	310.6	42.0]
B4D		siltstone	248/18W	0.05	7\7	349.4	-72.4	6.4	89	67.4	82.8
						[63.8	-89.3	7.4	68	296.0	63.2]
B4E		diabase plug		91.29	7\7	261.1	-72.8	25.2	--	--	---
			South Shetland Islands: Byers Peninsula, Beaches South of Point Smellie (298.85°E, 62.65°S)								
B5A		andesite dike	103/74W	29.48	5\6	15.3	-63.5	14.3	30	152.4	70.4
B5C		sandstone	204/23W	0.10	5\5	3.9	-67.2	3.9	380	130.3	77.1
						[56.9	-64.1	4.0	375	208.0	54.3
B5D		sandstone	214/21W	0.45	6\6	2.8	-69.0	3.7	329	128.5	79.8
						[61.6	-69.6	3.7	330	221.4	57.5]
B5E		sandstone	219/25W	0.34	5\5	5.7	-72.2	9.4	68	149.1	83.9
						[76.0	-71.1	8.7	78	234.1	52.7]
B5F		andesite dike	109/79W	49.00	7\7	21.5	-74.3	4.3	199	207.1	79.6
B5G		andesite dike	97/72W	19.08	5\5	351.0	-70.8	13.0	36	83.2	81.2
B5H		sandstone	234/26W	0.77	4\4	344.6	-70.2	15.4	37	68.2	78.5
						[100.5	-79.8	15.4	37	264.5	53.9]
B5H		andesite dike		3.22	3\3	340.7	-64.8	19.1	43	75.5	70.8
B5I		andesite dike	129/90	36.65	5\7	358.6	-63.7	10.4	55	115.5	72.7
			South Shetland Islands: Byers Peninsula, Sealer Hill (298.9°E, 62.67°S)								
B6A		diabase plug		134.61	6\6	31.8	-60.0	3.8	312	174.8	61.2
B6B		andesite plug		18.44	6\6	13.2	-70.1	5.1	174	163.5	79.0
B6C		diabase plug		252.05	7\7	62.0	-79.2	5.1	141	250.8	65.1
B6D		diabase plug		402.84	5\5	79.1	-74.4	3.3	534	242.7	54.9
B6E		diabase plug		152.70	6\7	34.3	-62.8	5.4	156	182.0	63.0

(Com A, B), component A or B; n\N, number of samples used for site mean calculation\total number of samples; D, declination; I, inclination; brackets indicate tilt-corrected data; a95, radius of 95% error circle at the 95% confidence level for site mean direction; k, estimate of precision parameter; VGP, virtual geomagnetic pole.

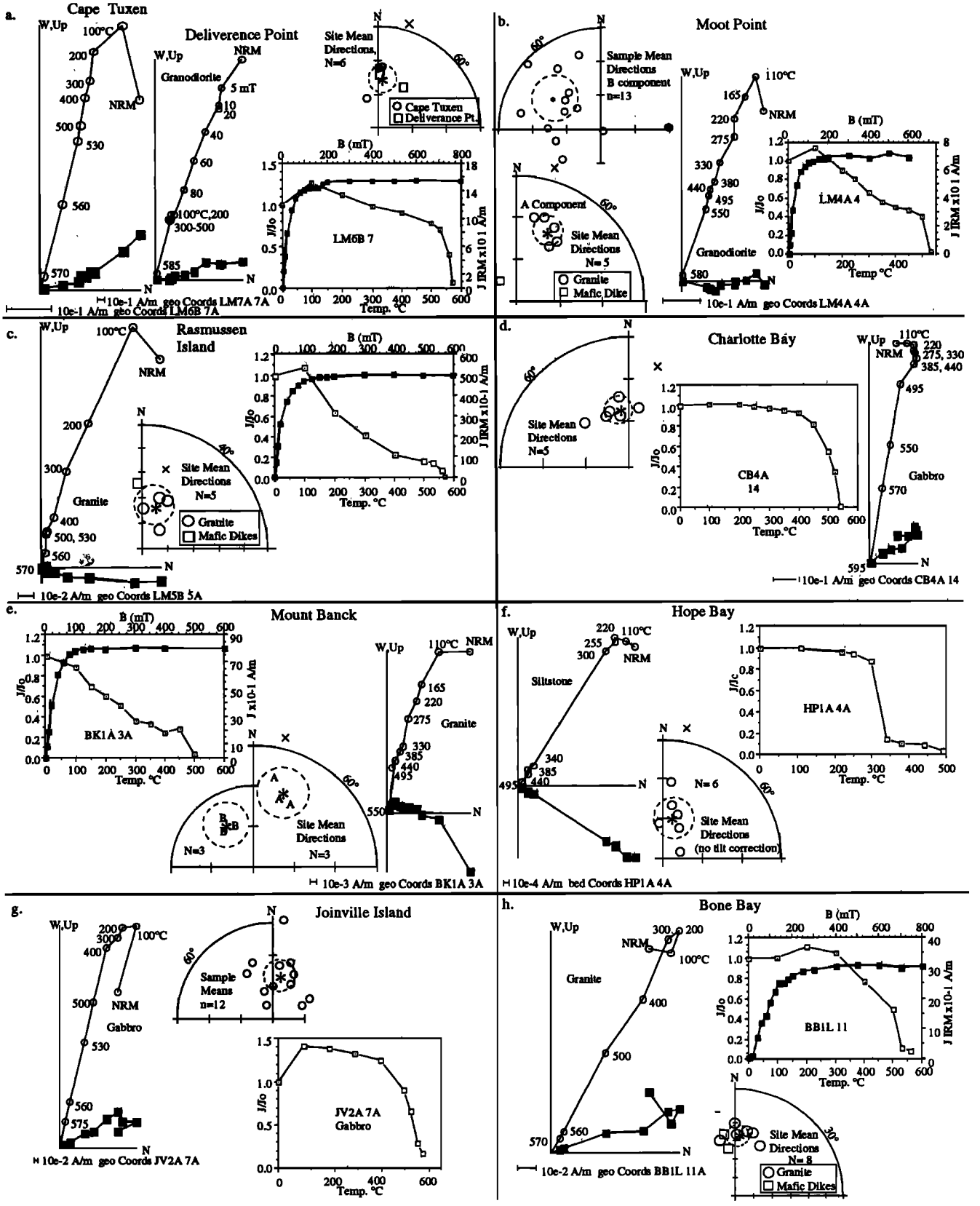


Fig. 2. Orthogonal projection of vector endpoint diagrams showing demagnetization behavior of samples from the Antarctic Peninsula. Open circles (solid circles) are projections on the vertical (horizontal) planes at the indicated levels of AF (mT, millitesla) or thermal (°C) cleaning. Stereographic projections of characteristic cleaned sample and/or site mean directions. Open (solid) symbols are on the upper (lower) hemisphere; crosses, present-day field; dashed circle, α95 radius of 95% confidence; asterisk, mean direction. Normalized intensity (J/Jo) plots are shown for localities where thermal demagnetization was used. IRM acquisition curves (solid squares) for representative samples are shown (sometimes with the intensity plots); B, direct inductance in millitesla; J IRM, IRM intensity.

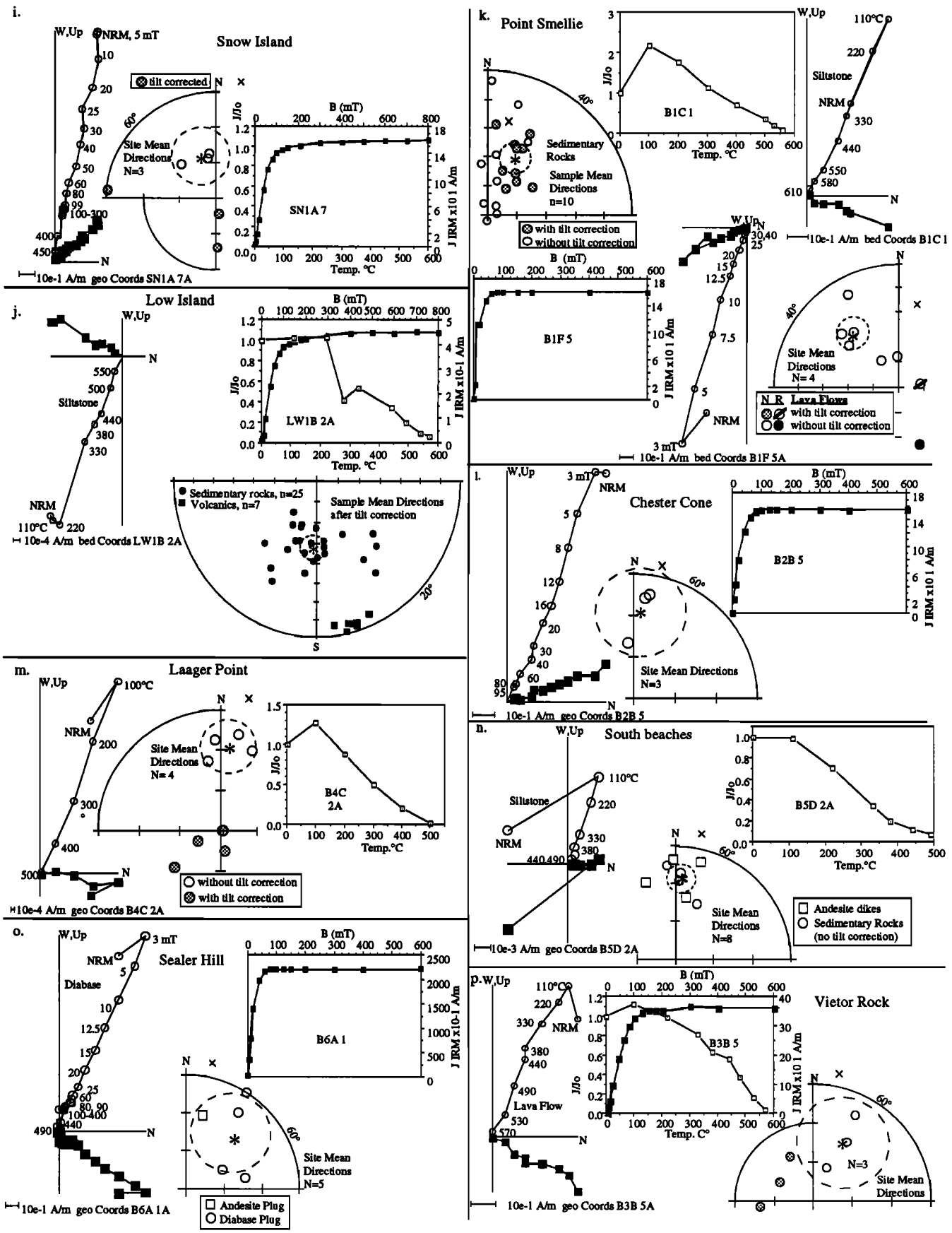


Fig. 2. (continued)

TABLE 2. Antarctic Peninsula Location Mean Directions and Paleopoles From This Study

Locality	Lithology	~Age*, Ma	N	Paleomagnetic Component	Polarity	Tests	D, deg	I, deg	α95	k	PPLon, °E	PPLat, °S	A95°	A63°	R	dp°, dm°	PLAT °S
Cape Tuxen	gabbro	85 ± 1	3		n		358.1	-75.3	8.7	201	64.1	87.2	15.4		2.9692		62
	granodiorite	84 ± 1	3		n		11.1	-74.9	7.2	298	175.1	83.8	13.0		2.9780		62
	combined site means		6				4.7	-75.2	4.2	260	152.3	86.4	7.5	3.7	5.9379		62
Moot Point	diorite and mafic dike	106 ± 1	5	A	n		23.8	-71.2	3.9	379	180.3	75.3	7.2	3.0	4.9646		56
		~55?	13†	B	n&r		297.2	-76.3	5.9	51	359.3	64.2				10.0,10.8	64
Rasmussen Island	granite	117 ± 1	4		n		21.5	-73.5	7.7	142	179.6	-79.7	14.5	5.8	3.9268		59
	granite and dike		5		n		14.4	-72.1	7.8	98	158.1	-79.9	13.2	6.6	4.8840		57
Charlotte Bay	gabbro, monzonite, and granite	?	5		n		349.3	-77.1	3.3	546	14.2	85.4	6.0	3.1	4.9754		65
Mt. Banck	granite	131 ± 4	3	A	n		21.6	-70.4	6.2	395	173.7	75.1	10.0	4.3	2.9871		54
			10†	B	n		327.1	-77.6	5.5	79	6.3	76.5				9.6, 10.2	66
Hope Bay	siltstones	?	6		n	- Fold	15.3	-79.4	5.1	172	275.3	79.1	7.1	4.7	5.9447		69
			6				[34.1	-64.4	12.0	32	183.2	67.5	15.3		5.7503		46]
Joinville Island	gabbro and dikes	~90?	12†		n		8.7	-77.0	5.1	73	245.0	85.8				8.9, 9.5	65
Bone Bay	granite	160 ± 1	6		n		9.9	-57.8		58	131.1	66.4	11.4		5.8602		39
	mafic dikes		2		n		357.5	-59.3		125							39
	combined site means		8				2.5	-57.3	5.9	89	125.4	64.5	8.2	4.4	7.8502		38
Snow Island	volcaniclastic rocks	~55?	3		n	- Fold	341.5	-77.5	8.1	230	357.1	81.4	15.1	6.5	2.9705		66
							[242.3	-80.0	31.5	16	321.1	50.4	56.0		2.6618		70]
Low Island	siltstones	Oxfordian	25†		r		208.9	49.9	9.9	10	161.1	52.6				8.8, 13.2	31
			25†			+ Fold	[183.3	58.0	6.0	24	123.9	65.3				6.5, 8.9	39]
Pt. Smellie	lava flows	Berriasian	4		n&r		327.3	-71.9	20.0	21	98.3	78.3	33.0		3.6574		57
			4		n&r	+Fold, Class C	320.1	-62.2	6.3	213	49.8	60.1	9.5	4.5	3.9685		44]
	siltstones	Valanginian-Hauterivian	3		n	+ Fold	9.0	-69.7	26.9	22	155.1	82.8	40.0			54	
			3				[24.5	-66.9	9.3	178	175.9	71.8	12.7	5.5	2.9790		50]
			11†		n		9.8	-74.2	9.7	23	188.2	84.8				15.8,17.5	60
			11†				[24.6	-69.2	4.9	90	172.8	73.7				7.0,8.3	53]
Chester Cone	andesite plug	?<110	3		n		4.7	-68.9	10.9	129	134.2	80.0	16.9	7.2	2.9631		52
Laager Point	sandstones	post - Barremian?	4		n		5.3	-69.0	6.7	190	136.3	79.7	11.0		3.9573		52
			4			50% Fold	{358.2	-81.7	3.3	790	303.9	77.2	5.7		3.9885		74}
			4				[210.5	-85.1	6.9	180	307.1	54.1	13.4		3.9374		80]
South Beaches	sandstones	post - Barremian?	4		n	- Fold	1.6	-68.8	4.7	391	116.4	81.2	7.4		3.9806		52
			4				[66.6	-70.7	6.9	176	232.2	56.4	16.0		3.9118		54]
	andesite dikes	?	4		n		6.1	-68.5	8.3	123	139.0	79.0	13.0		3.9411		52
	combined site means	?	8				4.2	-68.6	3.8	213	128.9	80.3	6.1	3.3	7.9158		52
Sealer Hill	diabase plug	~108 ± 4	4		n		45.6	-70.8	13.0	51	215.3	65.8	20.9	9.9	3.8523		55
	andesite plug	?	6†		n		13.2	-70.1	5.1	174	163.5	79.0				7.6, 8.8	54
	combined site means	~108	5				38.9	-70.6	10.8	51	208.6	68.9	17.7	8.6	4.7966		56
Vietor Rock	lava flows	107 ± 4	3		n		27.0	-73.0	11.8	111	206.5	76.6	19.0	8.1	2.9537		59
			3				[298.1	-79.4	11.7	112	331.3	63.2	29.9		2.8895		69]

N, number of sites used for calculation of mean directions and paleopole (sites listed in Table 1 with α95s > 15° have not been used in calculations); Component, component of magnetization; polarity: n, normal; r, reversed; paleomagnetic tests: - Fold, negative fold test, + Fold, positive fold test, class C, reversal category of McFadden and McElhinny [1990]; *D*, declination; *I*, inclination; brackets indicate tilt-corrected; braces indicate 50% unfolding; α95 error circle at the 95% confidence level; *k*, estimate of precision parameter; PPLon and PPLat, paleopole longitude and latitude; A_{95}, radius of error circle at 95% confidence level for the mean paleopole based on the site mean VGPs; A_{63}, radius of error circle at 63% confidence level; *R*, resultant vector; *dp°*, *dm°*, oval of 95% confidence about the pole position; PLat, paleolatitude. The data underlined are shown in Figure 4a, considered the best results for each location.

* See text for age assignments.

† Sample means used for paleopole calculation.

Antarctic Peninsula Volcanic Group [*Elliot and Gracanin*, 1983].

Forty two samples were collected from the Mount Flora Formation, representing seven sites. All samples were demagnetized using thermal techniques. The low unblocking temperature of the magnetization in the Mount Flora sedimentary rocks (75% of the magnetization lost between 300° and 360°C) and the pervasive pyritization could indicate that the magnetic carrier is pyrrhotite [*Lowrie*, 1990]. The Mount Flora Formation rocks do not pass a fold test [*McElhinny*, 1964]. The mean direction without tilt correction is D= 15.3°, I = -79.4°, N= 6 sites (Tables 1 and 2 and Figure 2f). The tilting, metamorphism, and pyritization of the rocks could be due to the emplacement of the overlying volcanic rocks or to the emplacement of a diorite intrusion (of unknown age) nearby to the sedimentary rocks.

Joinville Island. Sixty-seven samples from eleven sites were collected from the Mount Alexander promontory on the southern side of Joinville Island. Samples were collected in Nordenskjöld Formation siltstones, igneous dikes, and a biotite + pyroxene-bearing gabbro. The sedimentary rocks appear baked and silicified; they are cut by northwest-southeast trending dikes. The samples from the sedimentary rocks proved impossible to interpret as they contained multicomponent magnetizations that were inconsistent between samples and sites.

The gabbro intrudes the Nordenskjöld Formation at Mount Alexander (sites JV2). The mean directions for nine samples from the gabbro and three from an igneous dike from site JV1 are D= 8.7°, I = -77° (Table 2 and Figure 2g). Four of the gabbro samples (site JV2B) drilled near the contact with the sedimentary rocks were more silicic, finer-grained, and magnetically unstable.

Bone Bay. A pink, coarse-grained, biotite-bearing granite cut by north-south trending, nearly vertical set of comagmatic? mafic dikes crop out along the eastern side of Bone Bay. The sites are spread over >15 km interval. On the basis of the intrusive relationships (xenoliths of dike in granite and granite xenoliths in dikes), the mafic dikes appear to be slightly younger than the granite and were emplaced while the granite was not fully cooled. The granite has yielded a 160 ± 1 Ma Rb-Sr whole rock age (R.J. Pankhurst, personal communication, 1992).

Nine sites were sampled from the granite and mafic dikes, including baked contact tests in the zone immediately adjacent to the dikes. Given that the dikes appear to be comagmatic with the granite intrusion, it is not surprising that differences were not observed in the directions between the granite and mafic dike sites, representing a positive, but inconclusive baked contact test. The rocks yielded a single component of magnetization that is upward pointing to the north (Table 1). The mean direction for eight sites (seven in the granite and one from a mafic dike; one granite site was unstable) is D= 2.5°, I= -57.3° (Tables 1 and 2 and Figure 2h).

South Shetland Islands

Snow Island. Basaltic and andesitic flows occur above a Lower Cretaceous volcaniclastic sequence [*Askin*, 1983; *Smellie et al.*, 1984] and are intruded by sills which have yielded a 54 Ma 40Ar/39Ar whole rock age [*Watts et al.*, 1984]. Three sites were drilled in gently dipping volcanic and volcaniclastic layers at the eastern end of President Head.

The magnetization is single component and points upward to the northwest (Tables 1 and 2 and Figure 2i). One site in red (hematitic?) volcanic rocks contained a high Tub component between 620° and 660°C believed to be due to hematite, while the component of magnetization from two other sites, with an unblocking temperature of ~560°C, suggests titanomagnetite as the magnetic carrier (Figure 2i). The mean directions for the three sites from Snow Island are D= 341.5°, I= -77.5° and fail the fold test [*McElhinny*, 1964] (Table 2 and Figure 2i).

Low Island. Samples were collected on Low Island at Cape Wallace from volcanic and marine volcaniclastic rocks of Oxfordian age (Figure 1) [*Smellie*, 1980; *Thomson*, 1982; *Smellie et al.*, 1984]. The sedimentary rocks are crystal tuffs and volcanic claystones metamorphosed to very low grade due to either burial metamorphism or low-grade contact metamorphism [*Smellie*, 1980; *Smellie et al.*, 1984]. The rocks are tilted and are cut by numerous brittle high-angle to vertical faults (some with vertical slickensides and 0.5 to 1 m gouge zones). A granodiorite (K-Ar whole rock age of 121 ± 3 Ma) intrudes the sedimentary rocks at Cape Wallace [*Smellie et al.*, 1984]. No sites were drilled in the granodiorite because of pervasive brittle fracturing.

Samples were drilled in the sedimentary rocks at two sites on the eastern side of Cape Wallace and at one site in volcanic rocks (on an islet off the northwestern shore of Cape Wallace). The sedimentary rocks yield directions that point down and to the south and pass the fold test at the 95% confidence level [*McElhinny*, 1964]. The sample mean directions were determined between the 340°C and 550°C steps and produce a mean of D= 183.3°, I= 58° (n=25). The volcanic rocks produce shallow inclinations which could be due to an incorrect tilt correction (it was very difficult to determine layering) or to paleosecular variation.

Byers Peninsula. Samples were collected from Byers Peninsula (Figure 3), on the western end of Livingston Island (Figure 1b) from Berriasian-Valanginian (~140 to ~125 Ma using the Decade of North American Geology (DNAG) time scale [*Palmer*, 1983]) marine to nonmarine sedimentary rocks and volcanic flows, Lower Cretaceous intrusions and lava flows [*Smellie et al.*, 1980, 1984]. Heavy snow cover and poor weather conditions prevented us from sampling many locations, and the rocks visited were not well exposed except in canyons and, at low tide, as wave-cut platforms. Samples were collected at six main locations (Figure 3).

Point Smellie: Berriasian-Valanginian marine sedimentary rocks and intercalated andesitic lava flows crop out in an east-west trending syncline at Point Smellie [*Smellie et al.*, 1980, 1984]. The rocks occur in the upper part of the mixed-marine member [*Crame and Farquharson*, 1985]. Sites were collected in the lavas and in gently dipping, well-bedded, grey-green siltstones and sandstones on both limbs of the syncline. Seven sites were sampled at Point Smellie (four in the flows, two in the sedimentary rocks, and one in baked contact adjacent to one of the flows).

The flows were successfully demagnetized using AF techniques although thermal methods produced the same results. After one site of reversed directions is inverted 180° (site B1F), the flows yielded north and up directions (D= 320.1°, I= -62.2°, N=4 sites after tilt correction) (Tables 1 and 2 and Figure 2k). The mean directions of the sites pass a reversal test (class C) [*McFadden and McElhinny*, 1990] and the fold test at the 95% confidence level [*McElhinny*, 1964]. While the volcanic rocks are altered, the alteration state is

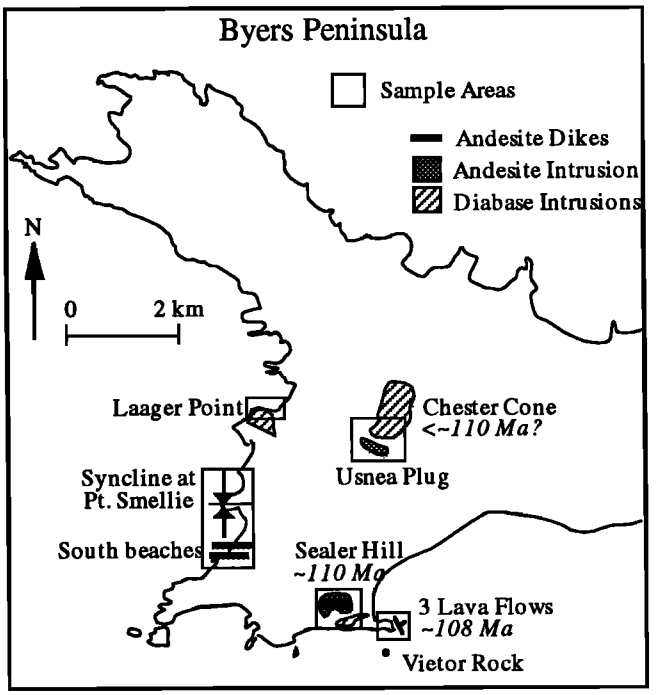


Fig. 3. Sample locations on Byers Peninsula.

related to their emplacement into wet sediments rather than to low-grade regional metamorphism.

The sedimentary rocks pass a fold test at the 95% confidence level [*McElhinny*, 1964] and yield directions of D= 24.5°, I= -66.9°, N=3 sites after tilt correction, which is not similar to the directions from the interlayered andesitic flows (Tables 1 and 2 and Figure 2k). It is difficult to explain why the magnetization in the sedimentary rocks is different from the lava flows except to speculate that one lithology has been completely remagnetized. Given the dual polarity nature of the flows and only normal polarity in the sedimentary rocks, it seems possible that the flows are yielding a primary magnetization and that perhaps the sedimentary rocks were remagnetized prior to folding. The time of folding is thought to postdate closely deposition of the rocks (D.H. Elliot, personal communication, 1992). Also, tilted but not folded Barremian (119-124 Ma) non-marine sedimentary rocks on Byers Peninsula [*Hernández and Azcárate*, 1971] rest unconformably upon the folded Berriasian-Valanginian strata which would suggest that the folding occurred prior to the Barremian but after the Berriasian-Valanginian deposition of the Point Smellie sedimentary rocks.

Chester Cone/Usnea Plug: Chester Cone consists of a hornblende-andesite plug or volcanic dome [*Pankhurst et al.*, 1979]. Usnea Plug, adjacent to Chester Cone, is a diabase intrusion of unknown age (Figure 3). Two sites were drilled at Chester Cone and one at Usnea Plug. The three sites produce a single north and upward pointing direction of D= 4.7°, I= -68.9° (Tables 1 and 2 and Figure 2l).

Isotopic dating of these rocks has proved difficult with K-Ar whole rock ages ranging between 107 ± 4 and 116 ± 5 Ma [*Pankhurst et al.*, 1979]. A rhyolitic flow which reportedly cuts the Chester Cone plug yields a Rb-Sr age of 111 ± 4 Ma [*Pankhurst*, 1982], but hornblende separates yield 40Ar/39Ar and K-Ar ages of ~130 Ma from the plug [*Pankhurst et al.*, 1979; *Watts et al.*, 1984] and may be due to excess argon or to a younger event resetting the K-Ar and Rb-Sr whole rock systems [*Smellie et al.*, 1984]. *Pankhurst et al.* [1979] observed hornblende breaking down to magnetite in the Chester Cone rocks and this might suggest that the magnetization is associated with the event that caused the hornblende alteration.

Laager Point: This location consists of Berriasian-Valanginian green-to-grey marine siltstones and sandstones cut by a doleritic plug (of unknown age) along a wave-cut platform (Figure 3) [*Smellie et al.*, 1984]. The sedimentary and structural relationships are complex due to localized extensive fracturing, faulting, and folding within the rocks.

The sedimentary rocks fail the fold test [*McElhinny*, 1964] possibly because of remagnetization associated with the deformation. The site mean direction, without tilt correction, is D= 5.3°, I= -69°, N=4 sites but the directions appear to be best grouped with a 50% tilt correction (D=358.2°, I=-81.7°) (Tables 1 and 2 and Figure 2m). Diabase samples collected within 10 m of the margin of the plug yielded stable but not internally consistent directions.

South Beaches: Shallow marine sedimentary rocks (Berriasian-Valanginian) crop out along the beaches south of Point Smellie (Figure 3) [*Smellie et al.*, 1984]. Plant-bearing grey and green sandstones, siltstones, and conglomerates, similar to the sedimentary rocks at Point Smellie, are gently warped in a dome and basin fashion [*Crame and Fahquahrson*, 1985]. East-west trending fractures, faults, and andesitic dikes of unknown age cut the folded sedimentary rocks. Individual dikes, generally 0.5 m to 2 m wide and sometimes zoned, appear to intrude along the axial surfaces of the gentle folds. Eight sites were drilled at this locality: five in andesite dikes and three in the sedimentary rocks. Baked contact tests were done on sedimentary rocks immediately adjacent to the dikes. No discernible difference was found between the magnetic directions in the sedimentary rocks, the baked sedimentary rocks and the andesite dikes (Tables 1 and 2), which could suggest that the sedimentary rocks had been remagnetized during emplacement of the dikes.

A single north and upward pointing component was found in the dikes and sedimentary rocks after thermal and AF techniques. The mean directions for both the dikes and sedimentary rocks are D= 4.2°, I= -68.6, N=8 sites (Tables 1 and 2 and Figure 2n).

Sealer Hill: Andesite and diabase intrusions occur at Sealer Hill (Figure 3) [*Smellie et al.*, 1984] and here, one site was drilled in the andesitic rocks and four in the diabase plug. The diabase yields ~108 ± 4 Ma K-Ar whole rock ages [*Pankhurst et al.*, 1979] and a 111 Ma $^{40}Ar/^{39}Ar$ age (no error given) [*Watts et al.*, 1984]. Both intrusions yield north and upward pointing directions. The mean directions from the diabase and andesite plugs are D= 38.9°, I= -70.6°, N= 5 sites (Tables 1 and 2 and Figure 2o). The site mean directions from the diabase fall into two general groups of two sites each (Table 1). The difference in directions between these two groups may be due to tilting, as each group (B6C and B6D; B6A and B6E) occurs on a different ridge which may represent different fault blocks around the volcanic dome. The mean directions are similar to those obtained here by *Watts et al.* [1984].

Vietor Rock: Three eastward dipping lava flows occur on the spit adjacent to Vietor Rock. The lava flows yield an average 40Ar/39Ar whole rock age of 104 Ma (no error given) [*Watts et al.*, 1984] while K-Ar whole rock ages average ~ 107 ± 4 Ma [*Pankhurst et al.*, 1979]. The lava flows yield a single component of magnetization that is up and to the north. The

mean direction for the three lava flows is D= 27°, I= -73° (Tables 1 and 2 and Figure 2p). The ~ 20° tilt of the lavas is thought here to be related to extrusion from the Sealer Hill volcanic dome (500 m away) and has not been tilt-corrected. The data are similar to those obtained from these same lava flows by *Watts et al.* [1984].

Samples collected from the Botany Bay Group sandstones and siltstones at Camp Hill (Figure 1) proved to be weakly magnetized and did not yield internally consistent directions.

INTERPRETATION OF ANTARCTIC PENINSULA DATA

All mean poles of the localities sampled in this study from the Antarctic Peninsula are shown in Figure 4a and underlined in Table 2. The paleopoles are shown in Figure 4a with their A_{63}s in order to better judge the differences between them. Consistency arguments would suggest that there has not been significant tilting in most of the AP block intrusive rocks since the magnetization was acquired because rocks of equivalent age but different lithologies sampled over a large area and across known structural boundaries yield similar results that are not the present-day field.

The AP block poles from several localities in this study have been combined to produce time averaged poles (each with an uncertainty of about ±5 m.y.) at ~155 Ma, ~140 Ma, ~130 Ma, ~110 Ma, and 85 Ma and are compared with previously published AP block poles (Table 3 and Figure 4b). As can be seen in Figure 4a, the paleopoles determined from the reversed polarity Low Island Oxfordian sedimentary rocks and the 160 Ma ± 1 Ma Bone Bay granite and mafic dikes (located 150 km across the Branfield Strait from Low Island) agree well with each other but not with any other AP block poles suggesting that the magnetization is likely to be primary (Table 3). The Low Island reversed polarity magnetization was probably acquired prior to the Cretaceous normal polarity interval since the data give paleolatitudes too low (40°) for a Late Cretaceous to Tertiary paleoposition unless Low Island was exotic. The site mean VGPs from the Bone Bay rocks and the Low Island sedimentary rocks produce a ~155 Ma pole of 124°E, 64°S, A_{95}= 7.1°, N=10 (Table 3 and Figures 4b).

The results from the Point Smellie lava flows of Berriasian-Valinginian age (131-144 Ma) are not like any of the other results from the Antarctic Peninsula (Table 2 and Figure 4a). The flows pass a fold test and are of both normal and reversed polarity. It is difficult to explain why the lava flow pole is so different from the pole determined from the intercalated sedimentary rocks except to suggest that the sedimentary rocks were remagnetized prior to folding.

The poles from the "A" component of the 131 ± 4 Ma Mount Banck granite (located 250 km from Point Smellie) and the Point Smellie sedimentary rocks (probable pre-Barremian and post- Berriasian/Valanginian magnetization age) have been combined to yield ~130 Ma pole of 182°E, 74°S, A_{95}= 5.9°, N=6 site mean poles (Table 3 and Figures 4a and 4b).

The Moot Point diorite (~106 Ma), the Vietor Rock lava flows (104-111 Ma) and the Sealer Hill diabase plug (~108 Ma) site mean VGPs have been combined to yield a ~110 Ma pole of 199.3°E, 73.8°S, A_{95}= 6.9°, K= 37 for N=13 VGPs (Table 3; Figure 4b). The ~110 Ma pole is similar to a pole (195°E, 65°S, A_{95}= 9°, N=11 VGPs) (Figure 4b) obtained from equivalent age rocks from the Orville Coast in southern Palmer Land (Figure 1) [*Kellogg and Rowley*, 1989].

The very close similarity between the ~110 Ma and 130 Ma AP poles from this study may be due to unrecognized remagnetization of the 130 Ma rocks but this seems unlikely given the different sample lithologies (granite, mafic dikes, siltstones), the geographically different locations (Gerlache Strait and Byers Peninsula; Figure 1), petrographic analysis and the positive fold test used to define the 130 Ma pole (Table 3). Perhaps the similarity in the 110 and 130 Ma AP poles reflects relatively little apparent polar wander for the AP block, a situation observed in the East Antarctic palemagnetic reference poles (Table 4). The Rasmussen Island granite (117 ± 1 Ma) and mafic dike produce a pole of 158.1°E, 79.9°S, A_{95}= 13.2°, N=5 VGPs which is not significantly different from the ~130 and ~110 Ma time-averaged poles supporting the idea of little apparent polar wander for the AP block during this interval (Figure 4a and 4b).

The site mean VGPs from the Cape Tuxen gabbro and Deliverance Point granodiorite produced an ~85 Ma mean pole of 152.3°E, 86.4°S, A_{95}= 7.5°, N=6 VGPs (Table 3; Figure 4b). The Cape Tuxen/Deliverance Point sites occur on both sides of a strike-slip fault, but no significant difference in directions was found between the sites to suggest tilting or rotation of directions from the faulting. In addition, similar results were obtained from a gabbro body ~10 km southeast of Cape Tuxen by *Blundell* [1962] suggesting that this is a large plutonic body that has not experienced discernible tilting across known faults. The Cape Tuxen pole is significantly different from one (81°E, 66°S) obtained from 4 sites in the 82 Ma Greenwich Island layered intrusive [*Watts et al.*, 1984] (Table 3 and Figures 1 and 4b) but agrees well with ~90 Ma East Antarctic and TI block paleopoles (Table 4 and Figure 5).

The Snow Island pole from volcaniclastic rocks is similar (although of higher latitude) to one obtained there from 54 Ma sills by *Watts et al.* [1984] (Table 3). The new Snow Island pole is indistinguishable from ~45 Ma and 50 Ma poles from Livingston and King George islands [*Dalziel et al.*, 1973; *Watts et al.*, 1984] (Figure 1 and Table 3).

Other poles shown in Figures 4a, that have not been used for new time-averaged AP poles include the Chester Cone/Usnea plug intrusions (<110 Ma?), the Laager Point sedimentary rocks, and the dikes and sedimentary rocks from the south beaches because of the uncertainty in the age of magnetization. The sedimentary rocks from Laager Point and the south beaches fail the fold test indicating remagnetization. The Chester Cone/Usnea plug rocks yield conflicting radiometric age data [*Smellie et al.*, 1984] and the andesite dikes from the southern beaches do not have radiometric age constraints. Chester Cone and the south beaches do, however, yield similar poles (Figure 4a) and are virtually identical to the ~95 Ma Byers Peninsula pole from *Watts et al.* [1984] (Table 3). Age dating has proved difficult from the Byers Peninsula igneous rocks as some of the rocks have experienced isotopic resetting caused by the emplacement of late hypabyssal intrusions 20 or 30 m.y. after cessation of extrusive volcanism [*Pankhurst et al.*, 1979; *Smellie et al.*, 1984]. The Chester Cone and south beaches poles are also similar to the 90 to 100 Ma AP block poles [*Kellogg and Reynolds*, 1978; *Valencio et al.*, 1979; *Watts et al.*, 1984] (Table 3). It would seem likely then, that the sedimentary rocks have been remagnetized and the andesitic dikes emplaced in the mid- to Late Cretaceous.

Results from the Charlotte Bay rocks (~115 ± 8 Ma from the monzonite; (Tangeman and Mukasa, personal communication, 1992)) indicate a very high latitude pole, unlike any poles from West Antarctica or East Antarctica except in the Late

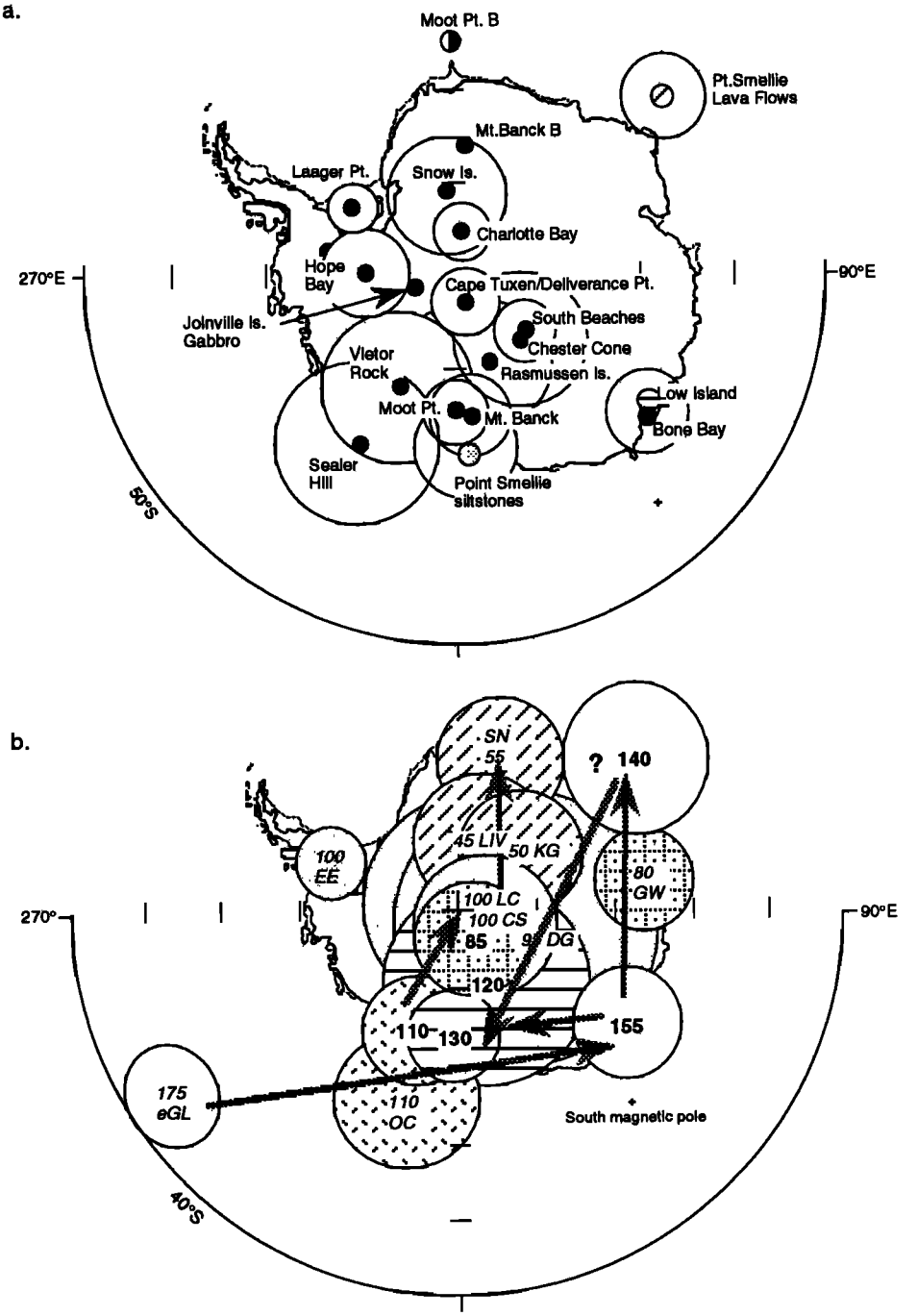


Fig. 4. (a) Equal-angle projection of AP block poles from this study with their respective A_{63} circles (except for those locations where paleopoles were calculated from sample mean directions rather than site mean virtual geomagnetic poles). Poles are from Table 2. Shaded circles, passes the fold test; half open circles normal and reversed polarity present; diagonal slashed circles, normal and reversed polarity and passes the fold test; horizontally lined circles reversed polarity and passes the fold test. (b) Time-averaged AP block poles (with A_{95} circles) from this study (in bold letters) with previously published AP block poles (in italics). See Table 3 for pole abbreviations. Paleopoles of similar age have been shaded and represent poles used to calculate the AP block time-averaged mean poles shown in Table 4 and Figure 5. Arrows indicate general trend of AP block apparent polar wander path.

Cretaceous and Cenozoic [*Watts et al.*, 1984]. The Charlotte Bay rocks may have been remagnetized due to emplacement of a large granitic intrusion exposed 6 km from the gabbro/monzonite location or the rocks may have been tilted as there is a substantial faulting and mineralization within a few kilometers of the sampling sites. The limited exposure allowed sampling from an area less than 50 x 50 m and could have been tectonically disturbed. The imprecision of the U-Pb zircon age from the monzonite which intrudes the undated gabbro, suggests either the presence of inherited zircons or Pb loss events, either of which would affect the interpreted age of the magnetization. For these reasons, the Charlotte Bay results have not been used for the tectonic reconstructions.

The B component (high Tub) of the 131 Ma Mount Banck granite yields a high latitude pole (Table 2) which falls into the Tertiary sector of the AP and East Antarctic APWP [*Watts et al.*,

TABLE 3. Mesozoic and Cenozoic Antarctic Peninsula Paleopoles

Time-Averaged Age, ± ~5 Ma	Age*, Ma	Longitude, °E	Latitude, °S	A95, deg	Location	Label on Figure 4b	Basis for Poles
~175	174 ± 5, 178 ± 2	237	46	6.4	East Graham Land †	*eGL*	*Longshaw and Griffiths* [1983]; lava flows, dikes, and granite
~155	160 ±1, Oxfordian	124	64	7.1	Byers Peninsula and northern Graham Land		N=10 site mean VGPs from Bone Bay (8) and Low Island (2)
~140		050	60	9.5	Byers Peninsula		N=4 site mean VGPs from Pt.Smellie lava flows
~130	131 ± 4, ~Valanginian	182	74	5.9	Byers Peninsula, Gerlache Strait		N=6 site mean poles from Mt. Banck (3) and Pt. Smellie (3)
~110	104 - 113	195	65	*9.0*	Orville Coast †	*OC*	*Kellogg and Rowley* [1989]; intrusions and dikes
~110	106 ± 1, 107 ± 4, 108 ± 4	199	74	6.9	Byers Peninsula, Penola Strait		N=13 site mean VGPs from Moot Point (5),Vietor Rock (3), and Sealer Hill (5)
~100	96 - 109	291	73	*4.6*	Eastern Ellsworth Land †	*EE*	*Kellogg and Rowley* [1989]; N=4 sites with a95s <15°
~100	95 - 110	074	87	*14.6*	Lassiter Coast †	*LC*	*Kellogg and Reynolds* [1978]; N=11 sites with a95s <15°
~100	94 ± 6, 102 ± 5	117	86	*9.0*	Gerlache Strait	*CS*	*Valencio et al.* [1979]; Cape Spring rhyodacite dikes and lava flow
?~95	75 - 111, average 96	132	77	*10.0*	Byers Peninsula		*Watts et al.* [1984]; plugs, sills and lava flows
?~95	?	136	80	4.8	Byers Peninsula		N= 11 site mean poles from Chester Cone and South beaches
~90	89 ± 11	106	81	*17.4*	Northern Graham Land	*DG*	*Watts et al.* [1984]; Danger Islands gabbro; age from *Hamer and Hyden* [1984]
?~90	?	162	81	*15.0*	Northern Graham Land		*Watts et al.* [1984]; Etna Islands gabbro
?~90	?	245	86	–	Northern Graham Land		n=13 samples from Joinville Island gabbro and dikes
~90	?	094	62	*4.0*	Northern Graham Land		*Watts et al.* [1984]; Wideopen Islands gabbro
~85	85 ± 1	152	86	7.5	Penola Strait		N=6 site mean VGPs from Cape Tuxen (3), Deliverance Point (3)
~80	82, 80 ± 2	081	66	*6.5*	South Shetland Islands	*GW*	*Watts et al.* [1984]; Greenwich Island intrusion; one age from *Smellie et al.* [1984]
~50	54	017	70	*8.2*	South Shetland Islands	*SN*	*Watts et al.* [1984]; Snow Island sills
?~50	?	357	81	15.1	South Shetland Islands		N=3 site-mean VGPs from Snow Island remagnetized siltstones
~50	49	048	79	*8.0*	South Shetland Islands	*KG*	*Watts et al.* [1984]; King George Island lava flows, plugs, and dikes
~50	46 ± 1	021	81	*9.0*	South Shetland Islands	*LIV*	*Dalziel et al.* [1973]; Livingston Island tonalite; isotopic age from *Smellie et al.* [1984]

All poles are shown in Figure 4b except those with question mark next to age. Italics indicate pole or A95 calculated by me (obtained from site mean directional data listed in the cited studies). See Table 1 for VGPs from this study.

*References for ages are cited in paleomagnetic studies unless mentioned specifically in text.

† See Figure 1a for location, all other locations on Figure 1b.

TABLE 4. Time-Averaged Antarctic Paleopoles in °E, °S (With Their A_{95}s)

Age, ± 5 Ma	East Antarctica	Antarctic Peninsula	Thurston Island	Ellsworth-Whitmore Mountains	Marie Byrd Land	References
175	220°, 55°, (3.9°)	237°, 46° (6.4°)		235°, 41° (5.3°)		1, 2
155	187°, 65° (6.9°) {1}	124°, 64° (7.1°)	145°, 64° (7°)			3, 4, 5
140	151°, 68° (3.8°) {2}	50°, 60° (9.5°)				4, 6
130	184°, 72° (5°)Δ {1}	182°, 74° (5.9°)	232°, 49° (7.9°)†			4, 7
110	198°, 76° (4°)# {2}	201°, 71° (8.1°)*	210°, 73° (7.6°)			4, 5, 8
100	179°, 80° (8.5°)# {2}	103°, 85°, (6°)*			242°, 65° (9°)¥	4, 8, 9
90	145°, 87° (8.4°)# {2}	152°, 86° (7.5°)	161°, 80°, (3.9°)			4, 5, 8
50	229°, 89° (10°) {3}	21°, 78° (7.5°)*				4, 10

References: 1, *Grunow et al.* [1987a]; 2, *Longshaw and Griffiths* [1983]; 3, *Schult and Guerreiro* [1979]; 4, *this paper*; 5, *Grunow et al.* [1991]; 6, *Van Fossen and Kent* [1993]; 7, *Grunow et al.* [1987b]; 8, *Hargraves* [1989]; 9, *Grindley and Oliver* [1983]; 10, *Embleton and McElhinny* [1982].

{1} Rotated to East Antarctica using *Norton and Sclater* [1979] and *Besse and Courtillot* [1988] rotation poles; {2} Rotated to East Antarctica using *Besse and Courtillot* [1988] rotation poles; {3} Rotated using *Royer and Sandwell* [1989] rotation poles.

Δ The average of the African Kaoko lavas pole [*Gidskehaug et al.*, 1975]; the South American Serra Geral SG1 pole [*Ernesto et al.*, 1990]; the North American Monteregian Hills [*Foster and Symons*, 1979] and Newfoundland dikes [*LaPointe*, 1979]. Kaoko lavas [see *Hargraves*, 1989] considered same age as Serra Geral lavas and related to South Atlantic Ocean basin opening.

† New radiometric data indicate average K-Ar biotite age of 126 ± 3 Ma for the TI block location [*Pankhurst et al.*, 1993].

¥ The rocks that define the MBL pole are between 90 and 110 Ma, the pole is listed under 100 Ma. # African reference poles are 110 Ma, Lupata volcanics; 100 Ma, Wadi Natash volcanics 1. For 90 Ma the combined pole means from Madagascar volcanics, Wadi Natash volcanics 2 and kimberlites KP1; N=3.

* See text and Table 3 for paleopoles used to calculate time-averaged AP block poles.

1984]. Eocene intrusions [*Scott*, 1965] which occur within the vicinity of Mount Banck could have caused the partial remagnetization. The high temperature nature of the B component magnetization may be due magnetite being precipitated from magmatic fluids produced during emplacement of younger plutons. Late stage magmatic fluids have caused alteration of feldspars, chloritization of biotite, noneutectic intergrowth of quartz and feldspar and the interstitial occurrence of quartz, calcite and epidote in the Mount Banck granite [*West*, 1974].

The B component of the Moot Point rocks is of both normal and reversed polarity. Since the rocks were emplaced during the Cretaceous quiet zone, the B component must be younger than 84 Ma. Within 10 km of Moot Point is a 55 ± 3 Ma quartz diorite [*Pankhurst*, 1982] that yielded D=144°, I= 82°, n=12 samples, $\alpha 95$=4° (reversed polarity) NRM directions [*Blundell*, 1962] very similar to the Moot Point B component. The B component pole is similar to the ~50 Ma Snow Island pole [*Watts et al.*, 1984]

The Hope Bay results have not been used in any interpretations because of the uncertainty in the age of the magnetization. The Mount Flora sedimentary rocks at Hope Bay fail the fold test and no estimate can be made of the time of magnetization. The Joinville Island gabbro yields a pole similar to the paleopoles obtained from the Danger Islands gabbro (89 ± 11 Ma [*Hamer and Hyden*, 1984]) and the undated Etna Islands gabbro (Table 3 and Figure 1b) [*Watts et al.*, 1984] and may be part of the same intrusive complex.

OROCLINAL BENDING, ROTATIONS OR APPARENT POLAR WANDER

The differences between the mid-Cretaceous poles from both the northern and southern Antarctic Peninsula (Table 3) can be interpreted as due either to relative tectonic movement or to apparent polar wander [*Kellogg*, 1980; *Watts et al.*, 1984; *Kellogg and Rowley*, 1989; *Grunow et al.*, 1991]. *Kellogg* [1980] suggested that the paleopole differences might reflect oroclinal bending of the southern Antarctic Peninsula, creating the southern part of the AP block "S" shape, or localized rotation of the Orville Coast area [*Kellogg and Rowley*, 1989]. This may be the case, but the suggestion by *Watts et al.* [1984] that the differences in AP block paleopoles reflect age differences and hence apparent polar wander is also possible. It appears from Figure 4b that the northern and southern ~110 Ma poles are similar to each other but are not similar to the 90 to 100 Ma AP block poles, supporting the idea of apparent polar wander in the mid-Cretaceous rather than oroclinal bending. In addition, there does not appear to be evidence for oroclinal bending of the northern Antarctic Peninsula since at least the Jurassic given the similarity of the EWM block 175 Ma and the TI block 155, 110 and 90 Ma paleopoles with equivalent age poles from the AP block (Table 4). Such good agreement of the AP, TI, and EWM block paleopoles at various times during the Mesozoic also supports the idea that there has been little tectonic tilting of the plutonic rocks from which most of the paleopoles were obtained.

The eastern Ellsworth Land pole [*Kellogg and Rowley*, 1989] is not similar to the other AP block poles (Figure 4b and Table 3) and this may be due to tectonic rotation along the Ellsworth Fault system (Figure 1a).

Using the assumption that there has not been oroclinal bending or tectonic rotation of the areas where the AP block paleopoles have been obtained (excluding the eastern

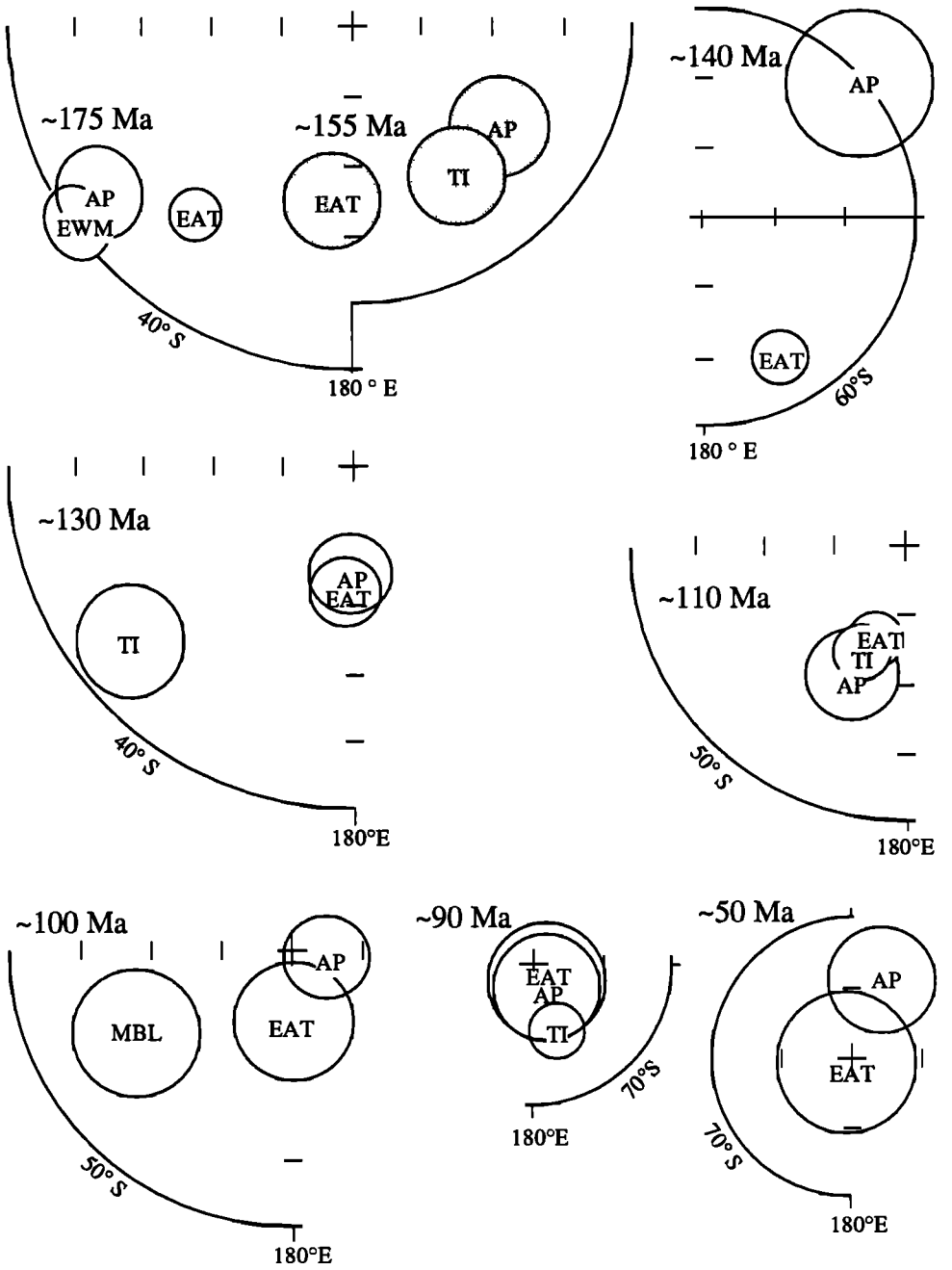


Fig. 5. Equal-angle projection of time-averaged paleomagnetic poles (with their A95s) for the West Antarctic crustal blocks (AP, TI, EWM, MBL) and East Antarctica (EAT); see Table 4 for poles.

Ellsworth Land pole), new AP block mean poles have been calculated for ~100 Ma and ~110 Ma from the poles listed in Tables 2 and 3. Location mean poles from the Sweeney Mountains and Hauberg Mountains on the Orville Coast [*Kellogg and Rowley*, 1989], Moot Point, Vietor Rock, and Sealer Hill have been averaged to produce a ~110 Ma AP block pole of 201°E, 71°S, A_{95}=8.1°, N=5. A ~100 Ma AP block paleopole (103°E, 85°S, A_{95}=6°, N=3 location mean paleopoles) was calculated from the Lassiter Coast [*Kellogg and Reynolds*, 1978], Cape Spring [*Valencio et al.*, 1979], and Danger Islands mean poles [*Watts et al.*, 1984] (Table 3). Finally, a ~50 Ma AP block paleopole (21°E, 78°S, A_{95}=7.5°, N=4 location mean poles) was calculated from the Livingston, King George, and Snow islands mean poles listed in Table 3.

COMPARISON OF PALEOMAGNETIC REFERENCE POLES

The paleomagnetic poles from the West Antarctic crustal blocks and East Antarctica are shown in Table 4 and Figure 5. Apart from the 175 Ma East Antarctic pole [*Grunow et al.*, 1987a], the other poles for East Antarctica have been obtained by rotating reference poles from other continents into an East Antarctic frame of reference from South America for 155 Ma [*Schult and Guerreiro*, 1979]; North America for ~140 Ma [*Van Fossen and Kent*, 1993]; Africa/North America/South America for ~130 Ma (see Table 4); Africa (Lupata volcanics) for 110 Ma; Africa (Wadi Natash volcanics 1) for 100 Ma, and Africa (Wadi Natash volcanics 2, Madagascar volcanics and kimberlites, KP1) for 90 Ma [*Hargraves*, 1989]; Australia for

50 Ma [*Embleton and McElhinny*, 1982]. All rotations to East Antarctica were performed using the *Norton and Sclater* [1979], *Besse and Courtillot* [1988], or *Royer and Sandwell* [1989] rotation parameters (see Table 4). The DNAG time scale [*Palmer*, 1983] has been used for correlation of the rotation poles with the paleomagnetic poles.

The new AP block poles have also been compared with other paleopoles from the West Antarctic blocks including the 175 Ma AP and EWM block poles [*Longshaw and Griffiths*, 1983; *Grunow et al.*, 1987a], the TI block ~150 Ma pole (shown with the ~155 Ma poles) [*Grunow et al.*, 1991], the TI 125 Ma pole (shown with the 130 Ma poles) [*Grunow et al.*, 1987b] (Table 4 and Figure 5). The 122 ± 2 Ma age of the rocks used for the TI block pole listed by *Grunow et al.* [1991] has been revised to a minimum age of ~125 Ma [*Pankhurst et al.*, 1993].

RECONSTRUCTIONS

A series of reconstructions have been made using the new AP block and other Antarctic poles (Table 4) for post-175 Ma, ~155 Ma and ~130 Ma. The rotation poles used to restore the AP and TI blocks to a position consistent with the paleomagnetic poles are listed in Table 5; rotation poles for the EWM and MBL blocks are listed by *Grunow et al.*, [1991]. The present continental outlines and subice topography have been used to define the West Antarctic crustal blocks in the reconstructions since the configuration of the blocks prior to Cenozoic time is unknown. The East Antarctic reference pole is used as the geographic south pole in Figures 6a-6c. The MBL block has been left in a mid-Cretaceous position [*Grindley and Oliver*, 1983] (Table 4) for the reconstructions because paleomagnetic constraints are not presently available for time periods prior to the mid-Cretaceous.

TECTONIC MODELS

Post ~175 Ma

The ~175 Ma Gondwanaland pre-breakup configuration of the West Antarctic blocks can be found in *Grunow et al.* [1991]. The new ~155 Ma paleomagnetic pole from the AP block suggests that the AP block rotated clockwise relative to East Antarctica after ~175 Ma and before ~155 Ma, reflecting early opening in the Weddell Sea and perhaps Somali and Mozambique basins (Figure 6a) [*Grunow, 1*993]. Although the oldest magnetic anomaly in the Somali-Mozambique basins is anomaly M25 (156 Ma) [*Simpson et al.*, 1979; *Cochran*, 1988], more recent analysis suggests that seafloor may extend back as far as 170 Ma [*Lawver et al.*, 1991]. Jurassic and Early Cretaceous marine magnetic anomalies from the Weddell Sea are problematical because there is little agreement on the age of the anomalies. The same anomalies have been identified as M29, M25, and M10 [*LaBrecque and Barker*, 1981; *La Brecque*, 1986; *Martin and Hartnady*, 1986].

Up to ~1000 km of Weddell Sea floor may have been created behind the southern AP and TI blocks between 175 Ma and ~155 Ma (Figures 6a and 6b) [*Grunow*, 1993]. The Gondwanaland (and hence East Antarctic) reference pole for ~155 Ma is not well constrained, however, and the amount of Weddell Sea oceanic crust created during the 175-155 Ma time interval will be dependent on the choice of reference pole.

TABLE 5. Finite Rotation Poles Relative to East Antarctica

Plate	Time, Ma	Longitude, deg	Latitude, deg	Angle, deg
Antarctic	175	-93.20	-72.30	35.48
Peninsula	155	141.18	16.98	16.86
	130	79.74	77.91	-16.42
Thurston	155	-22.82	29.04	-16.63
Island	130	102.09	74.58	-31.12

Positive rotations are counterclockwise when viewed from above the pole. °West longitude and south latitude are shown with minus.

~155 Ma

The new AP block paleomagnetic pole from Mount Banck and Byers Peninsula indicates that the AP block was near its present-day position relative to East Antarctica by 130 Ma. To reach that position, the AP block rotated counterclockwise between ~155 and ~130 Ma (Table 4 and Figure 6b). Counterclockwise rotation of the AP block may have been related to Weddell Sea opening switching to the northern half of the Antarctic Peninsula and into the Rocas Verdes marginal basin of southern South America [*Dalziel*, 1981; *Barker and Lonsdale*, 1991] while opening ceased in the southern AP block (Figure 6b) [*Grunow*, 1993].

The overall setting shown in Figure 6b predicts subduction of earlier formed Weddell Sea ocean floor beneath the southern AP and TI blocks due to the southward motion of an EWM block attached to East Antarctica [*Grunow*, 1993]. The deformation associated with the Palmer Land orogenic event along the southeastern AP block [*Kellogg and Rowley*, 1989] is attributed here to subduction of Weddell Sea ocean crust beneath the AP block (Figure 6b) [*Grunow*, 1993]. Small remnants of Jurassic ocean floor may still exist in the southwestern Weddell Sea basin and perhaps in the Byrd subglacial basin (Figure 1a).

~130 Ma

The TI block paleomagnetic data require clockwise rotation relative to the AP block and East Antarctica between ~130 Ma and ~100 Ma [*Grunow et al.*, 1987b]. To accommodate the TI block rotation, the opening geometry of *Lawver et al.* [1985] has been modified in the western Weddell Sea basin (Figure 6c).

The tectonic boundary between the AP and TI blocks may have been located at the base of the AP block in what is known as the Ellsworth Fault System (EFS) (Figure 1a) [*Kellogg and Rowley*, 1989]. On the basis of the AP and TI paleomagnetic data, the EFS was probably active in the Early Cretaceous marking the boundary between the AP and TI blocks. Faults with dominantly dextral offsets cut folded Jurassic and Lower Cretaceous rocks of the southern AP block [*Kellogg and Rowley*, 1989], and this deformation could be due to the dextral shearing of the EWM and TI blocks against the AP block in the Early Cretaceous. The model shown in Figure 6c also predicts sinistral strike-slip motion in the northern AP block related to initiation of sea-floor spreading in the South Atlantic region and cessation of spreading in the Rocas Verdes marginal basin in South America.

~110 Ma

The paleomagnetic poles from the AP and TI blocks from 110 Ma onward are similar to the East Antarctic reference poles (Figure 5) and suggests that these blocks were in or near to

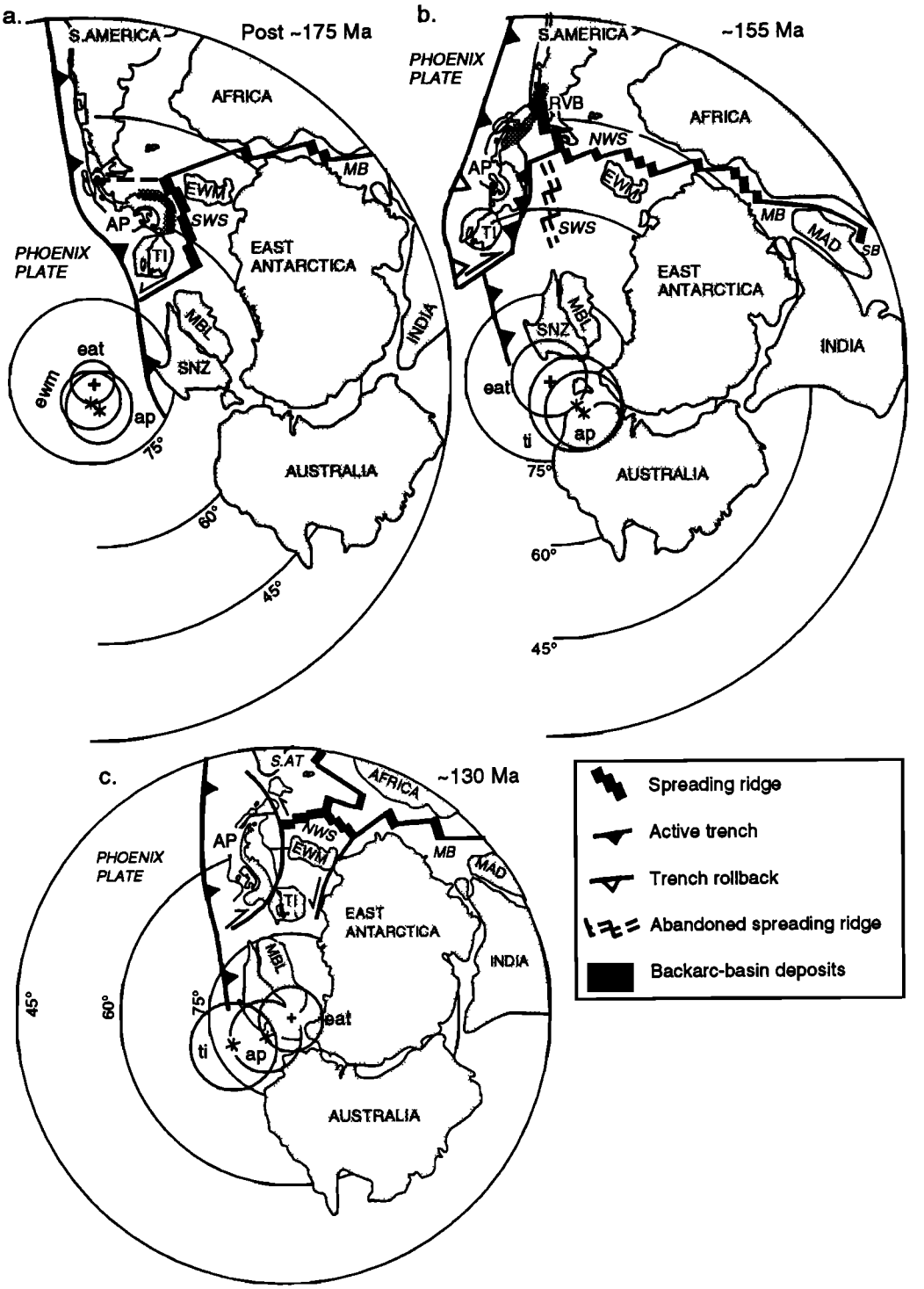


Fig. 6. Geographic south pole (+) is centered on the East Antarctic (eat) reference pole with A95 circle; asterisks mark West Antarctic paleopoles with A95 circles, ap=AP block, ewm= EWM block, ti=TI block. Abbreviations as in Figure 1 plus: AP = Antarctic Peninsula block; MAD=Madagascar; MB=Mozambique Basin; NWS = northern Weddell Sea; RVB=Rocas Verdes Basin; S.AT = South Atlantic ocean basin; SB=Somali Basin; SNZ=southern New Zealand; SWS=southern Weddell Sea. (a) Post-175 Ma. Initiation of seafloor spreading in Weddell Sea. Clockwise rotation of AP block relative to East Antarctica due to opening in Weddell Sea. Opening trends in Weddell Sea consistent with M25 magnetic anomalies in Mozambique basin. (b) ~155 Ma. Spreading geometry changes in Weddell Sea such that opening begins along northern AP block and into Rocas Verdes basin. Ocean floor in southern Weddell Sea begining to be subducted beneath southeastern AP and TI blocks. Counterclockwise rotation of AP block causing rollback along Phoenix-AP block trench. Creation of convergent margin along southeastern AP block causes Palmer Land event. (c) ~130 Ma. AP block near its present-day position with respect to East Antarctica. Change in Weddell Sea spreading geometry related to opening in South Atlantic ocean basin while opening ceases in Rocas Verdes basin. Dextral strike-slip motion in southeastern AP block related to movement of EWM and TI blocks. Sinistral strike-slip motion of ~750 km between EWM block and East Antarctica expected [*Grunow et al.*, 1991]. Sinistral motion expected along northern AP block as South America moves westward relative to AP block with opening in South Atlantic basin.

their present-day positions with respect to East Antarctica (Figure 1a). Recent multichannel data from the northern Antarctic Peninsula suggest that there has been no Cenozoic motion between the Antarctic Peninsula and the western Weddell Sea [*Barker and Lonsdale*, 1991], whereas Chron 34 clearly links the eastern Weddell Sea ocean floor with East Antarctica [*Lawver et al.*, 1991]; that is, there is no marine geophysical evidence for Cenozoic motion between the Antarctic Peninsula and East Antarctica.

CONCLUSIONS

The new ~155 AP block paleomagnetic pole from the Low Island sedimentary rocks and the Bone Bay granite suggests clockwise rotation relative to East Antarctica between 175 Ma and 155 Ma which may be connected with early opening in the Weddell Sea. At ~155 Ma, a change in the seafloor spreading direction may have occurred such that opening ceased along the southern Antarctic Peninsula and continued along the northern Antarctic Peninsula into the Rocas Verdes basin in South America. The ~130 Ma AP block and TI block poles suggest that these blocks rotated counterclockwise relative to East Antarctica between ~155 and 130 Ma . The AP and TI block rotation, in conjunction with the southward motion of East Antarctica during opening of the Somali-Mozambique-Weddell Sea basins is believed here to have resulted in subduction of Jurassic Weddell seafloor beneath the southern AP and TI blocks.

The AP-EWM-TI blocks appear to have been in or near their present-day positions with respect to East Antarctica by ~100-110 Ma. Similarity in northern and southern AP block mid-Cretaceous poles does not support oroclinal bending to explain the "S" shape of the Antarctic Peninsula.

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